Marginally Useful

Formalizing the Information Gap in Conformal Prediction

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Abstract

Conformal prediction gives finite-sample, distribution-free *marginal* coverage for a set. The guarantee is real, and it is often misread as evidence of forecast quality. We separate the two with one decomposition, which we call the residual-information gap: for a single-shape residual predictive system, the log-score regret relative to the oracle is exactly the mutual information $I(R; X)$ between the residual and the input. Conformalization re-levels coverage but cannot touch this quantity, because it is a property of the predictor's shape class and not of calibration; no recalibration that ignores X reduces it. The familiar cautions about conformal prediction then follow as context rather than discoveries: marginal coverage is not conditional (and distribution-free conditional coverage is impossible without structure), validity is insensitive to sharpness, and the guarantee requires exchangeability. On a coverage–score plane, conformalizing a fixed model is a purely horizontal move, toward correct coverage and never toward a better score. The same decomposition marks the converse: when coverage itself is the objective, the gap is irrelevant and conformal prediction is exactly right. All figures are reproducible.

1 Introduction

A practitioner maintains a probabilistic forecaster and is told, repeatedly, that conformal prediction is the principled, distribution-free way to "add uncertainty."[1] They bolt it on. The number they actually report (a proper score such as the predictive log-likelihood) does not improve. The natural diagnosis is a tuning failure. The real one is a *category error*: conformal prediction is a method for *certifying the coverage of a set*, and it is being asked to *estimate a distribution*. These are different kinds of object, graded by different instruments, and no amount of tuning turns one into the other. None of this is a criticism of the method; it is a guide to where the method belongs.

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[1] The reading is pervasive, and not only in blogs. Applied and survey writing routinely calls the marginal-coverage guarantee "well-calibrated" or "reliable" uncertainty quantification (the word *calibrated* standing in for coverage (Bellotti and Zhao, 2025)), and tutorial expositions go further, presenting conformal prediction as a way to "add uncertainty" (tds, a), the one thing you need for it (tds, b), or even to "predict full probability distributions" (Manokhin). The core *methodological* literature is more careful (it separates *validity* from *efficiency*), and a growing critique strand makes the present point independently: that valid-but-uninformative sets have little practical value (Mehrtens et al., 2025), and that interval length can be gamed while marginal coverage is preserved (Min et al., 2026): the same trivial-validity point, in the coverage–length setting, that we revisit in Section 3 and that is itself long-standing folklore. Our quarrel is with the conflation, not with conformal prediction.

None of this impugns the mathematics. Given exchangeable data, a fitted predictor, and a nonconformity score, split conformal prediction returns a set $C_\alpha(x)$ with

$$\mathbb{P}\big(Y_{n+1} \in C_\alpha(X_{n+1})\big) \;\geq\; 1-\alpha, \tag{1}$$

finite-sample and with no assumption on the data distribution (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023). The trouble is entirely in the reading of (1). The probability averages over X_{n+1} as well as Y_{n+1}: it is a statement about the *average* future input, not about the one in front of you. It is *marginal* coverage, and as the foundational analyses warned from the start, "a good estimator must satisfy something more than marginal coverage" (Lei and Wasserman, 2014).

What is new, and what is not. This paper has one new result. For a single-shape residual predictive system, the log-score regret to the oracle equals the mutual information $I(R;X)$ between the residual and the input, a quantity that no recalibration ignoring X can reduce; we call it the residual-information gap (Section 5, Proposition 3). It says exactly what conformalization leaves on the table, and why.

Everything else is scaffolding for that statement, and is labelled accordingly. The impossibility of distribution-free conditional coverage (Section 3) is the theorem of Lei and Wasserman (2014) and Foygel Barber et al. (2021); we restate it and claim none of it. That coverage cannot constrain the log-score (Section 4) is folklore, recorded here as a precise warm-up. The coverage–score plane (Section 6) is a presentation device, not a result. The remaining sections place the gap in context: exchangeability and time series (Section 7), the converse when coverage really is the objective (Section 8), and the scope against the modern literature (Section 9), each advance of which buys its gains by the very conditional modeling the gap identifies.

2 Setup: split conformal and the marginal guarantee

Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable. Fix a point predictor $\hat{\mu}$ trained on a disjoint set, and a *nonconformity score* $s(x,y)$; the canonical choice in regression is the absolute residual $s(x,y) = |y - \hat{\mu}(x)|$. Compute calibration scores $s_i = s(X_i, Y_i)$ for $i = 1, \dots, n$ and let

$$\hat{q} \;=\; s_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil, \tag{2}$$

the k-th order statistic (with $\hat{q} = +\infty$ when $k > n$). The split-conformal set is

$$C_\alpha(x) = \{y :\; s(x,y) \leq \hat{q}\} \;=\; [\hat{\mu}(x) - \hat{q},\; \hat{\mu}(x) + \hat{q}] \tag{3}$$

for the absolute-residual score. Exchangeability of the augmented scores makes the rank of s_{n+1} uniform, which yields (1) (Vovk et al., 2005; Lei et al., 2018). The construction is simple and the guarantee is real, and one readily verifies that empirical coverage tracks $1-\alpha$ as the sample size, noise level, and α are varied. None of this is in dispute; the question is what it is taken to mean.

3 Three known limits of the marginal guarantee

None of the three facts below is new. We restate them, with attribution, to fix notation and to mark where the contribution of Section 5 departs from what is already established.

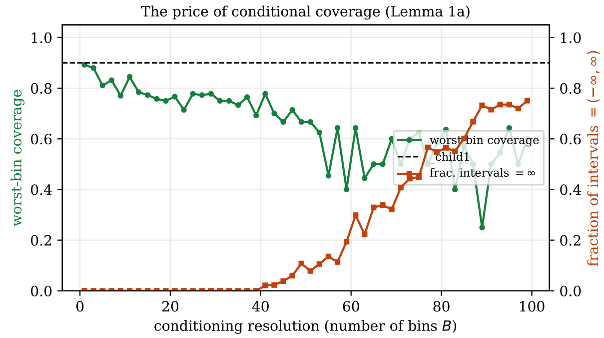

Figure 1: The price of conditional coverage (Lemma 1a). As the conditioning resolution B grows, per-cell calibration sets shrink and the only valid cell interval becomes $(-\infty, \infty)$; uniform (worst-bin) coverage is bought only as the fraction of infinite-length intervals rises.

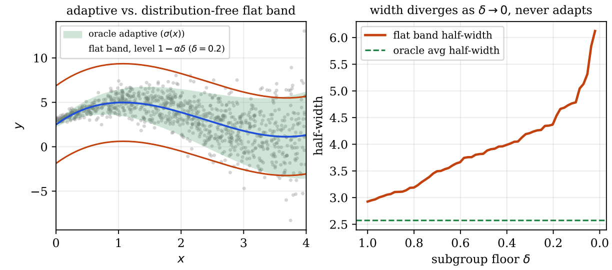

Figure 2: Subgroup coverage buys only a wider band (Lemma 1b). The distribution-free protection of all δ-subgroups is a flat, non-adaptive band at level $1 - \alpha\delta$ whose width diverges as $\delta \to 0$; the adaptive oracle that uses $\sigma(x)$ is far tighter but requires the distributional assumption.

(i) Marginal is not conditional, and conditional is impossible for free. One would prefer *conditional* coverage, $\mathbb{P}\big(Y \in C(x) \,|\, X = x\big) \geq 1 - \alpha$ for (almost) every x. Distribution-free, this cannot be bought. Lei and Wasserman (2014) show (their Lemma 1) that any band with non-trivial finite-sample conditional validity has *infinite* expected length at almost every point of a continuous distribution. Foygel Barber et al. (2021) restate and sharpen this: a conditionally valid C_n has $\mathbb{E}\,\mathrm{leb}(C_n(x)) = \infty$ at almost all non-atomic x (their Prop. 2.2), and even the relaxed demand of coverage on every subgroup of probability $\geq \delta$ "is impossible to attain beyond the trivial solution" of inflating the marginal level to $1 - \alpha\delta$ (their Thm. 3.1). We collect both as the operative no-go result.

Lemma 1 (No-go for distribution-free conditional coverage)**.** *Let P have non-atomic X. (a) (Lei and Wasserman, 2014; Foygel Barber et al., 2021) Any C_n with non-trivial finite-sample conditional validity has $\mathbb{E}\,\mathrm{leb}\big(C_n(x)\big) = \infty$ at almost every x. (b) (Foygel Barber et al., 2021) Requiring $\mathbb{P}\big(Y \in C_n(X) \,|\, X \in G\big) \geq 1 - \alpha$ for every measurable G with $P(X \in G) \geq \delta$ is attainable only by a C_n whose marginal level is inflated to $1 - \alpha\delta$, a uniformly wider, non-adaptive band.*

These are not loose bounds to be tightened; they are the reason the gap cannot be closed for free, and both surface as concrete finite-sample facts. Part (a) is illustrated in Figure 1: localizing split conformal to recover per-x coverage shrinks each cell's calibration count n_b until $\lceil (n_b+1)(1-\alpha) \rceil > n_b$, at which point (2) gives $\hat{q} = +\infty$ and the only valid cell interval is $(-\infty, \infty)$, the infinite expected length of the lemma, arriving as arithmetic, so that uniform coverage is recovered only as the fraction of infinite-length intervals grows. Part (b) is Figure 2: the distribution-free protection of all δ-subgroups is exactly the flat band at level $1 - \alpha\delta$, whose width diverges as $\delta \to 0$ while never adapting to x, at a multiplicative width cost over an assumption-driven adaptive oracle.

So the guarantee you can have distribution-free is the marginal one, and a marginally valid band "tends to overestimate the set when x is in the high density area and to underestimate for low density x" (Lei and Wasserman, 2014): it over-covers the easy inputs and under-covers the hard ones, silent by construction about the cases the model was built to get right. Figure 3 exhibits exactly this on heteroscedastic data: 90% overall, near-100% where the noise is small, well below target where it is large.

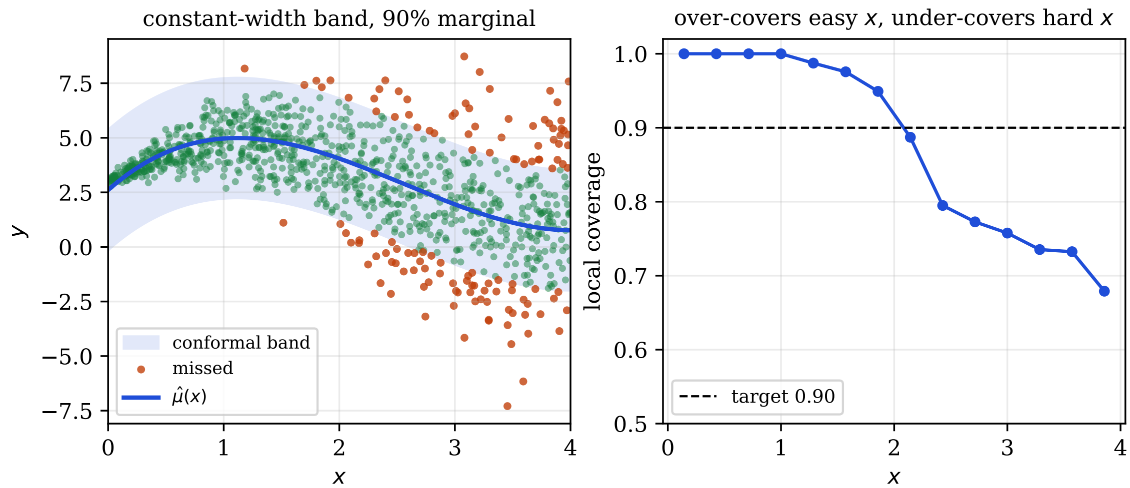

Figure 3: Marginal is not conditional. A constant-width split-conformal band (left) attains 90% *marginal* coverage on heteroscedastic data, but its *local* coverage (right) runs near 100% where the noise is small and well below target where it is large: it over-covers the easy inputs and under-covers the hard ones.

To be clear, the conformal literature is not unaware of this distinction. The standard vocabulary already separates *validity* from *efficiency* and adaptivity, and modern methods such as conformalized quantile regression (Romano et al., 2019) deliberately use better conditional modeling to obtain more adaptive sets. The narrower claim here is that the distribution-free marginal coverage *certificate* should not be mistaken for distributional forecast quality, and that, as the next sections make precise, whatever adaptivity such methods gain is bought by conditional modeling, not by the conformal step itself.

(ii) Validity is trivially satisfiable. A predictor that returns the whole outcome space with probability $1-\alpha$ and the empty set otherwise satisfies (1) exactly and conveys nothing. More practically, the *same* coverage attaches to an excellent $\hat{\mu}$ and to a deliberately terrible one; the latter merely yields a wider $\hat{q}$ in (2). Marginal validity is therefore not evidence of a good method; it certifies the fence, not the cattle. The next section makes this precise.

(iii) Without exchangeability, even the marginal guarantee fails. Equation (1) rests on exchangeability of the augmented sample. Under covariate shift, label shift, or temporal dependence it does not hold, and the patches that restore something do so by weakening what is guaranteed (Section 7).

4 Coverage and score are orthogonal

We first make "coverage is not a measure of forecast quality" a precise, constructive fact. Model a forecaster as a pair $(f(\cdot \mid x), s)$ of a predictive density and a nonconformity score. The point is conditional on the score: it is sharp when the density is *not* used to build s (as for the canonical residual score), and it does not apply to density- or quantile-based scores.

Proposition 1 (Coverage does not constrain the log-score)**.** *The split-conformal set* (3) *is a measurable function of the nonconformity scores* $\{s_i\}$ *and of* $s(x, \cdot)$*; it is not a functional of the predictive density* f *unless* f *is explicitly used in the score. Consequently, if two forecasters share the same score function* s *and the same calibration scores, their conformal sets, and hence their marginal coverage at every level* α*, are identical, while their expected log-scores* $\mathbb{E}[\log f(Y \mid X)]$ *may differ by an arbitrary amount.*

Proof. The first claim is immediate from (2)–(3): $\hat{q}$ is an order statistic of $\{s_i\}$ and $C_\alpha(x) = \{y : s(x, y) \le \hat{q}\}$, so two forecasters agreeing on s and on $\{s_i\}$ produce the same set for every x, α, and (1) concerns only that set.

For the gap, take X degenerate and $Y \sim N(0, 1)$; let $\hat{\mu} \equiv 0$ with the absolute-residual score $s(x, y) = |y|$, which does not depend on the predictive density at all. For any claimed predictive density $f_\sigma = N(0, \sigma^2)$ the scores are $|Y_i|$, independent of σ, so the conformal set is the fixed interval $[-\hat{q}, \hat{q}]$ with $\hat{q}$ the empirical $(1 - \alpha)$-quantile of $|Y|$. Yet

$$\mathbb{E}[\log f_\sigma(Y)] \;=\; -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{\mathbb{E}[Y^2]}{2\sigma^2} \;=\; -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \;\xrightarrow[\sigma \to 0^+]{}\; -\infty,$$

and likewise $\to -\infty$ as $\sigma \to \infty$. Hence for any $M > 0$ there are σ_1, σ_2 with identical conformal sets (identical marginal coverage at every α) yet log-scores differing by more than M. $\square$

Corollary 1. *For a residual-score conformal predictor, marginal validity certifies an order statistic of residual magnitudes. It does not identify, bound, or rank the scale, shape, or sharpness of any predictive density that was not used to construct the score; in particular, no inference about forecast quality (as graded by a strictly proper score) can be drawn from coverage alone.*

This is the formal version of a point several authors have made informally: that conformal validity "almost invites you to use garbage prediction functions" (Recht, 2024). Figure 4 shows it: the claimed density's spread slides freely while the conformal interval and its 90% coverage do not move, because the spread never entered (2).

5 The residual-information gap

This section is the paper's contribution; the rest of the paper is setup for it and consequences of it. Forecast quality decomposes into *calibration* and *sharpness*, and the discipline's stated goal is to maximize sharpness subject to calibration, assessed by a proper scoring rule (Gneiting et al., 2007; Gneiting and Raftery, 2007). Within the class of objects conformal prediction can produce, it is log-score optimal; the loss is the class.

Conformal predictive systems. A conformal predictor can be made to output a distribution rather than a set, the *conformal predictive system* (CPS) of Vovk et al. (2019), built from *signed* residual scores. It is the mode in which conformal prediction estimates a full distribution, so it is the form the criticism must address. We must, however, distinguish it from the more common practice of reading the nested symmetric intervals of ordinary *absolute*-residual split conformal as if they were a distribution; the two re-level different objects, and the distinction matters for the gap below.

Proposition 2 (Re-leveling, two cases)**.** *Fix a location predictor* $\hat{\mu}$ *with residual* $R = Y - \hat{\mu}(X)$ *of marginal density* g *and CDF* G*. In the large-calibration limit (under exchangeability):*

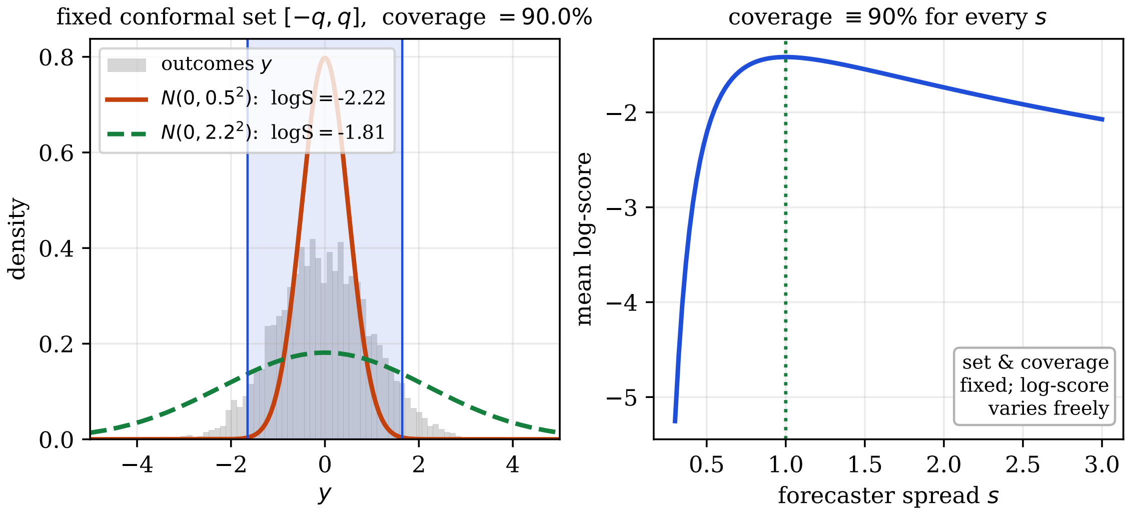

Figure 4: Coverage does not constrain the log-score (Proposition 1). With a residual score the conformal set $[-q, q]$ is fixed by the outcomes alone (left), so two predictive densities with very different log-scores share the *same* set and the same 90% coverage. Sweeping the forecaster's spread s (right) moves the log-score freely while coverage stays pinned at 90%.

(A) Signed-residual CPS. The CPS predictive distribution is $\Pi_x(y) = G\big(y - \hat{\mu}(x)\big)$ (the calibration ECDF $\widehat{G}_n \to G$ by Glivenko–Cantelli), with density $\pi_x(y) = g\big(y - \hat{\mu}(x)\big)$: the base location forecast re-leveled by the marginal signed-residual law.

(B) Absolute-residual intervals. Ordinary split conformal at level α returns the symmetric interval $\hat{\mu}(x) \pm \hat{q}_\alpha$, with $\hat{q}_\alpha$ the $(1-\alpha)$-quantile of $|R|$. Read as a predictive distribution (by sweeping α) it is symmetric about $\hat{\mu}(x)$ with $|\cdot|$ distributed as $|R|$; its density is the symmetrized marginal residual law $h_{\text{sym}}(z) = \frac{1}{2}\big(g(z) + g(-z)\big)$.

Proof sketch. (A) The CPS at level α returns endpoints equal to the signed-residual order statistics; sweeping α traces $\hat{\mu}(x)$ plus the residual quantile function, i.e. $\widehat{G}_n^{-1}$ shifted by $\hat{\mu}(x)$, so $\Pi_x(y) = \widehat{G}_n(y - \hat{\mu}(x))$; take limits (Vovk et al., 2019). (B) The intervals depend on the data only through $|R|$ and are symmetric about $\hat{\mu}(x)$; the unique symmetric density whose absolute value has the law of $|R|$ (with signed marginal g) is $h_{\text{sym}}(z) = \frac{1}{2}(g(z) + g(-z))$. □

In both cases the estimation was done by the base model and the residual sample; conformal supplies only the leveling. We now quantify the cost. Write the true conditional residual density as $r(\cdot \mid x)$, so the oracle density is $q^\star(y \mid x) = r\big(y - \hat{\mu}(x) \mid x\big)$, and let $\bar{r} = \mathbb{E}_X\, r(\cdot \mid X) = g$ be the marginal residual density.

Proposition 3 (Residual-information gap)**.** *Fix the location predictor $\hat{\mu}$ and write $I(R; X) = \mathbb{E}_X\, \mathrm{KL}\big(r(\cdot \mid X) \,\|\, \bar{r}\big) \geq 0$, the mutual information between the residual and the input. The expected log-score regret relative to the oracle $q^\star$ is*

$$\textit{(A) signed CPS:} \quad \mathbb{E}[\log q^\star(Y \mid X)] - \mathbb{E}[\log \pi_X(Y)] \;=\; I(R; X), \tag{4}$$

$$\textit{(B) absolute intervals:} \quad \mathbb{E}[\log q^\star(Y \mid X)] - \mathbb{E}[\log h_{\text{sym}}(R)] \;=\; I(R; X) + \mathrm{KL}\big(\bar{r} \,\|\, h_{\text{sym}}\big). \tag{5}$$

Both are ≥ 0. The term $I(R; X)$ vanishes iff $R \perp X$; the extra term $\mathrm{KL}(\bar{r} \,\|\, h_{\text{sym}})$ vanishes iff the marginal residual law is symmetric. Among all single-shape forecasters $y \mapsto h(y - \hat{\mu}(x))$, $\mathbb{E}[\log h(R)]$

is maximized at $h = \bar{r}$; the signed CPS attains this optimum. Hence no recalibration that ignores X can reduce $I(R;X)$: it can at best replace the shape by $\bar{r}$, removing the skewness term in (5) *and reducing case (B) to case (A). Reducing $I(R;X)$ itself requires conditioning on X.*

Proof. With $R = Y - \hat{\mu}(X)$, $\log q^\star(Y \mid X) = \log r(R \mid X)$. (A) With $\pi_X(Y) = g(R) = \bar{r}(R)$, $\mathbb{E}[\log r(R \mid X) - \log \bar{r}(R)] = \mathbb{E}_X \mathbb{E}_{R \mid X} \log \frac{r(R \mid X)}{\bar{r}(R)} = \mathbb{E}_X \mathrm{KL}(r(\cdot \mid X) \,\|\, \bar{r}) = I(R;X)$, the definition of mutual information since $\bar{r}$ is the marginal of R. (B) Add and subtract $\log \bar{r}(R)$: $\mathbb{E}[\log r(R \mid X) - \log h_{\mathrm{sym}}(R)] = I(R;X) + \mathbb{E}_{R \sim \bar{r}} \log \frac{\bar{r}(R)}{h_{\mathrm{sym}}(R)} = I(R;X) + \mathrm{KL}(\bar{r} \,\|\, h_{\mathrm{sym}})$. For the optimum, $\mathbb{E}[\log \bar{r}(R)] - \mathbb{E}[\log h(R)] = \mathrm{KL}(\bar{r} \,\|\, h) \geq 0$ (Gibbs), so $h = \bar{r}$ is best among shared shapes; a single shape cannot represent X-dependence in r, so it cannot change $I(R;X)$. □

Remark 1 (What this says, and does not say)**.** The right-hand side of (4) is exactly the information about the residual distribution that remains in X after the location predictor is fixed. Calling this the residual-information gap makes the content of "blind to sharpness" precise. Three caveats. (1) *Conformal is not dominated within its class.* The signed CPS is log-score optimal among single-shape location forecasters; the cost is the restriction, not the calibration step. The absolute-residual interval system is the one that pays the extra $\mathrm{KL}(\bar{r} \,\|\, h_{\mathrm{sym}})$, for discarding sign information. (2) *What recalibration can and cannot do.* A recalibration that ignores X (e.g. marginal PIT recalibration) can move the shape toward $\bar{r}$ and thereby erase the skewness term, i.e. turn case (B) into case (A); it cannot touch $I(R;X)$. Reducing $I(R;X)$ requires modeling conditional residual shape: heteroscedastic models, conditional recalibration (Kuleshov et al., 2018), or conformalized quantile regression (Romano et al., 2019), which is the conformal world's way of leaving the single-shape class. (3) *The distribution-free guarantee is real.* Recalibration offers calibration only in expectation or asymptotically and carries no coverage certificate; conformal's marginal guarantee is finite-sample and assumption-light. When the certificate itself is the deliverable, that is decisive (Section 8).

Relation to prior observations. Both ingredients are individually known. That a conformal predictive system re-levels the marginal residual law is due to Vovk et al. (2019); that expected log-score regret decomposes into Kullback–Leibler terms is standard, and an information-theoretic reading of conformal prediction has been developed by Correia et al. (2024). The step taken here is the combination: identifying the regret of the single-shape system as exactly the conditional residual information $I(R;X)$, and reading off that no recalibration ignoring X can reduce it. The contribution is naming the quantity conformalization cannot touch, not the components it is assembled from.

Figure 5 draws the gap: under heteroscedasticity the true conditional residual law $r(\cdot \mid x)$ varies with x, while the single-shape system uses the marginal $\bar{r}$ everywhere, and the regret is exactly the information $I(R;X)$ that separates them. The special case $I(R;X) = 0$ ($\hat{\mu}$ correct and r a fixed Gaussian with the wrong scale) is the one where variance recalibration fit to the log-score recovers the shape and improves the score, while conformalization re-levels to exact coverage and returns a set.

6 The coverage–score plane

The preceding results suggest a single diagnostic, provided we are careful about what is being plotted. Define a *reporting system* as a set-valued report $C_\alpha(x)$ and, optionally, a density-valued

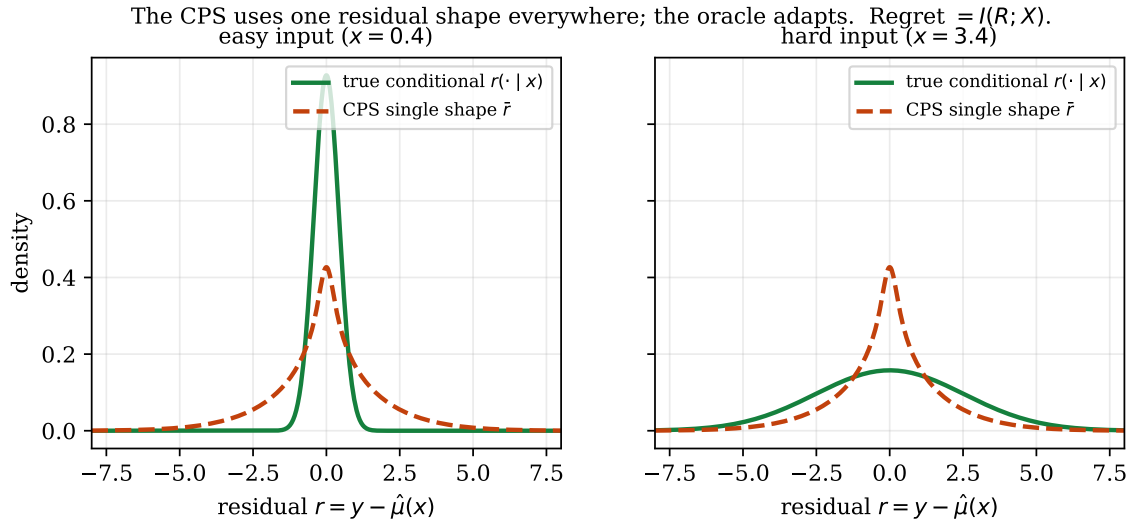

Figure 5: The residual-information gap (Propositions 2–3). A single-shape conformal predictive system uses the marginal residual law $\bar{r}$ for every input; the oracle uses the true conditional $r(\cdot \mid x)$, narrow for easy inputs (left) and wide for hard ones (right). The expected log-score regret is exactly the mutual information $I(R; X)$, which no X-ignoring recalibration can reduce.

report $f(\cdot \mid x)$. Its coordinates are

$$\big(C,\, S\big), \qquad C = \Big|\, \mathbb{P}(Y \in C_\alpha(X)) - (1-\alpha) \,\Big|, \qquad S = \mathbb{E}[\log f(Y \mid X)], \tag{6}$$

the marginal-coverage error of the *set* and the proper-score performance of the *density*. The decomposition of forecast quality into calibration and sharpness (Gneiting et al., 2007) is, in this plane, the statement that the two coordinates are distinct, and our results read geometrically:

- Residual split conformal is a horizontal projection. It maps a report $(C_\alpha^{\text{base}}, f)$ to $(C_\alpha^{\text{conf}}, f)$: it changes the set so that $C \to 0$ while leaving the density f (and hence S) untouched (Proposition 1). It moves a point left, never up.
- Conformal predictive systems are different: they supply their own density-valued report, so they do have a vertical coordinate, but that coordinate is governed by the single-shape restriction of Proposition 3, capped at the oracle minus $I(R; X)$.
- Recalibration and better modeling move vertically. Fitting shape to a proper score raises S; conditioning on x is the only way to raise it past the ceiling set by the residual-information gap (4).

A system reported by its coverage alone is reported by one coordinate of a two-coordinate quality. The plane is a presentation device (adjacent to reliability and sharpness diagrams (Gneiting et al., 2007)), but the "conformal = move left, never up" reading makes the category distinction hard to miss. Figure 6 populates the plane with the time-series methods of Section 7, plotting worst-window coverage against the interval (Winkler) score.

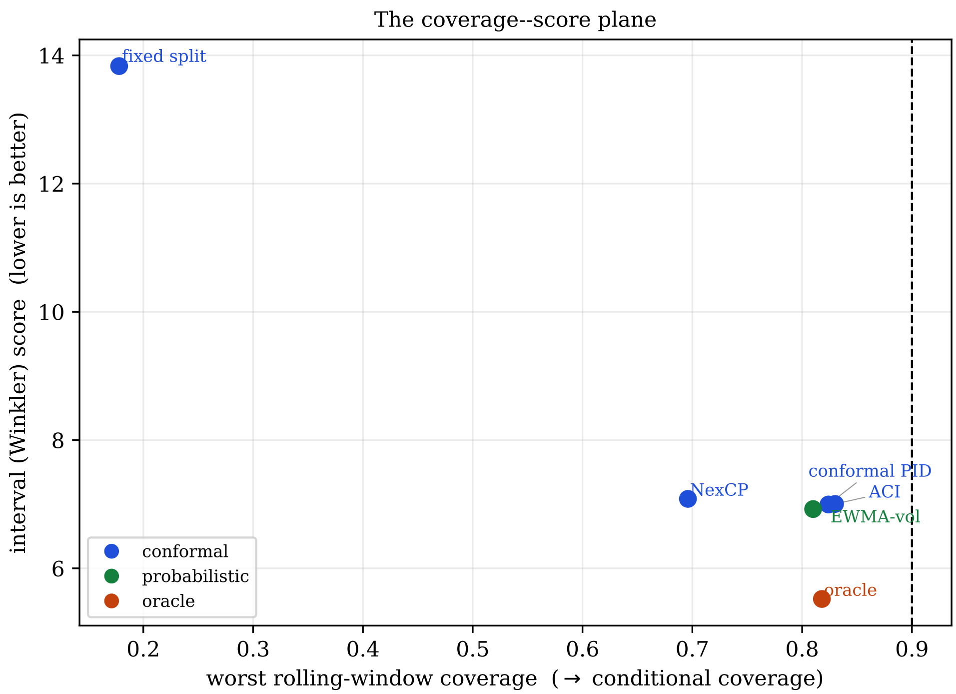

Figure 6: The coverage–efficiency plane, populated by the time-series methods of Section 7 (means over five seeds). The horizontal axis is worst rolling-window coverage (a proxy for conditional coverage); the vertical axis is the proper interval score. Coverage and forecast quality are distinct coordinates: the probabilistic volatility model and the oracle dominate on score, and no method reaches per-step conditional coverage.

7 Exchangeability and the time-series case

Drop exchangeability and (1) is no longer guaranteed. The literature's repairs are ingenious, and uniform in one respect: they survive by redefining what is guaranteed.

- Weighted conformal prediction (Tibshirani et al., 2019) restores marginal coverage under covariate shift when the likelihood ratio is known; with estimated weights coverage becomes approximate and the effective sample size drops.
- Adaptive Conformal Inference (Gibbs and Candès, 2021) updates the level online and "puts no constraints on the data generating distribution," but guarantees a *long-run time-average* miscoverage frequency, $\frac{1}{T}\sum_t \mathrm{err}_t \to \alpha$, not a finite-sample statement about $\mathbb{P}(Y \in C(X))$, and certainly not conditional coverage.
- EnbPI (Xu and Xie, 2021) drops exchangeability for approximate marginal coverage under strong-mixing/stationarity surrogates; later online methods (Zaffran et al., 2022; Gibbs and Candès, 2024) retreat to regret bounds over local intervals. The general beyond-exchangeability analysis of Barber et al. (2023) bounds the coverage gap by a total-variation distance that is exactly what nonstationarity inflates.

It is worth separating three distinct guarantees that the word "conformal" now spans: (i) *split conformal under exchangeability* gives finite-sample marginal coverage; (ii) *adaptive conformal inference* gives long-run, time-average coverage control under distribution shift; (iii) *online conformal*

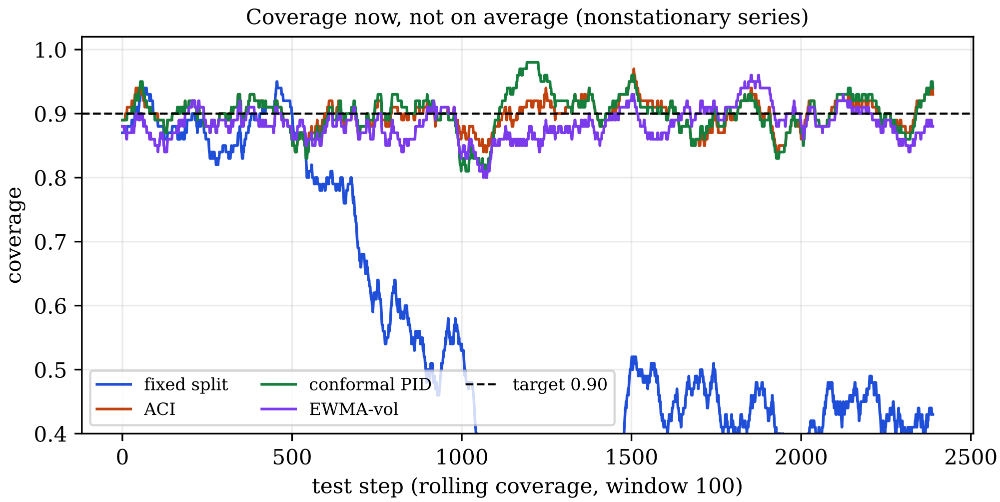

Figure 7: Coverage now, not on average. Rolling coverage on a nonstationary series: fixed split conformal collapses once the volatility regime arrives, while adaptive conformal (ACI, conformal PID) and a probabilistic EWMA-volatility model track the target. Even the recovering methods oscillate around the target rather than holding it pointwise.

under arbitrary shifts gives regret-style or local-interval performance rather than the original exchangeable guarantee. These are progressively weaker and more diffuse objects; none is the sharp predictive distribution *now*, for *this* step, that a proper score grades. The exchangeability required by guarantee (i) is typically unavailable in deployed time-series settings, though blocking or learned-transform constructions can sometimes recover an exchangeability-like assumption.

An experiment bears this out (Figure 7). On a nonstationary series with known σ_t, fixed split conformal collapses far below target once drift sets in, but the adaptive methods (ACI, conformal PID control, and weighted/NexCP) recover coverage and are competitive on the interval (Winkler) score. They are not straw men. The finding is the one the theory predicts: a simple probabilistic volatility model (an EWMA-variance Gaussian) matches or beats them on the proper score *and* returns a full distribution, and where the conformal methods help they do so by adapting the interval to recently realized residuals, that is, by local conditional modeling, with the conformal step supplying the coverage leveling on top. (Full tables, including MAPIE's EnbPI/ACI and `crepes`, are in the benchmark accompanying the paper.)

The same holds for an off-the-shelf probabilistic forecaster. The author's `skaters` (Cotton, 2021), a streaming forecaster that emits a predictive mean and standard deviation online, has the best non-oracle interval score (6.57) and CRPS (0.86) here, because its predicted spread tracks the changing volatility. The forecaster is deliberately simple (the `thinking_fast_and_slow` skater composes a fast moving average for the level with a slow moving average of the residuals for the spread), which sharpens the point: even a trivial adaptive model that estimates conditional spread beats every conformal method here, and conformalizing it cannot help. A normalized conformal wrap re-levels its coverage to $1-\alpha$ at an essentially identical interval score (6.58 against 6.57), and a naive split-conformal wrap on the nonstationary series collapses (to 60% coverage, score 11.9). The conformal step re-levels; it does not add sharpness, which is the experience the introduction describes.

8 When coverage *is* the objective

The argument is skeptical, not dismissive, and it comes with a precise converse. The litmus is one question:

> *Is your loss a function of whether the truth lands in a region, or of where it lands?*

If the former, coverage is the native object and conformal prediction is sound and often ideal:

- Selective prediction and risk control. "Return a small set guaranteed to contain the label 95% of the time; a human checks it." The deliverable is a set; risk-controlling prediction sets and conformal risk control formalize this (Angelopoulos et al., 2021; Bates et al., 2021; Angelopoulos et al., 2024).
- Retrieval and shortlisting, where the product is a candidate set and coverage is recall.
- Anomaly and novelty detection via conformal p-values, conformal prediction's most natural home, because here it is not a forecaster but a distribution-free hypothesis test and coverage is Type-I error control (Vovk et al., 2005).
- Compliance and long-run frequency control, where a certificate of the form "contained 95% of the time on average" *is* the product, and the distribution-free, finite-sample nature of (1) is exactly the value added.

Two caveats temper even these. If you trust a calibrated posterior you can form a sharper set yourself (the smallest region of mass $\geq 1 - \alpha$) and recover the expectations a bare set discards; conformal's marginal value is then purely the distribution-free guarantee, useful when you distrust the density and exchangeability holds. And even when coverage is the goal you usually want it conditionally, which returns us to the impossibility of Section 3.

9 Scope, and the constructive escapes

Our strong claims target the object that is actually over-sold: *post-hoc, marginal, split conformal prediction with a residual score.* A large and active literature builds conformal methods that do more, and how they relate matters, because in every case they confirm the paper's mechanism rather than contradict it: they buy sharpness or conditional coverage by *conditioning on X or optimizing a proper objective*, which is exactly what Propositions 3 say is required.

Conformal predictive systems that condition on X. Mondrian / binned conformal predictive systems (Boström et al., 2021) fit a separate residual law per region, and Toccaceli (2026) chooses the bins by minimizing a leave-one-out CRPS. These produce sharp, shape-adaptive predictive distributions, and they do so by leaving the single-shape class of Proposition 2 (a per-bin shape is X-dependent) and, in the CRPS case, by making the partition criterion a proper score. This is the most pointed case for the defence, and the concession is real: when the reference class is itself selected by a proper scoring rule, the conformal *construction* is performing conditional density estimation. The category distinction does not vanish; it relocates to "which component estimates the distribution," and the answer is the conditional partition, not the coverage certificate: the value is in the estimation, as throughout. Conformalized quantile regression (Romano et al., 2019) is the same story with a quantile model in place of the bins.

End-to-end conformal training. The orthogonality of Proposition 1 is a statement about *post-hoc* conformalization of a fixed predictor. Conformal training (Stutz et al., 2022; Correia et al., 2024) deliberately differentiates through the conformalizer to minimize expected set size subject to coverage, which couples the coverage and efficiency axes by construction. This does not contradict the proposition; it illustrates its corollary: to obtain sharpness you must optimize for it, rather than read it off the coverage certificate.

The information in set size. Correia et al. (2024) show that the expected size of a calibrated conformal set lower-bounds the conditional entropy $H(Y \mid X)$. This is complementary to our Proposition 3: both say conformal *geometry* carries information. It also sharpens the two-coordinate plane of Section 6: the *efficiency* (set-size / sharpness) coordinate is the information-bearing one, while the marginal *coverage* coordinate, by Proposition 1, is not. Reporting only the latter discards exactly the coordinate that encodes aleatoric uncertainty.

Approaching conditional coverage. The no-go result of Section 3 concerns *exact, distribution-free, finite-length* conditional coverage. It is not the end of the subject. Diagnostics such as the excess-risk-of-target-coverage family (Braun et al., 2025) quantify and decompose the conditional gap, and post-processing via pivotal scores and optimal transport (Laplante, 2026) attains *approximate* conditional coverage by estimating the conditional law of the score, again conditioning on X, and explicitly under the structural assumptions the impossibility theorem rules out for the exact, assumption-free case. These approach the goalpost; they do not move the one the no-go result defends.

Modern time-series conformal. Beyond the adaptive methods benchmarked in Section 7, methods such as HopCPT (Auer et al., 2023) exploit temporal structure to tighten intervals and improve empirical adaptivity. Their guarantees remain of the approximate / long-run kind catalogued there; the improvement, again, comes from modeling the conditional error law.

Beyond the log-score. We used the log-score for the clean information-theoretic identities, but the coverage–score plane and the argument carry over to any strictly proper score; the accompanying benchmark in fact ranks methods by the interval (Winkler) score and CRPS (Gneiting and Raftery, 2007), on which the same orderings hold.

10 Conclusion

Conformal prediction certifies the coverage of a set, finite-sample and distribution-free, under exchangeability. That guarantee is genuine and, for the right objective, exact. The error is reading it as distributional forecast quality. For a fixed location predictor the guarantee you can have (marginal) is rarely the one you want (conditional); the conditional one is unavailable distribution-free without further structure; the output is a set, not a measure; its log-score loss relative to the oracle is exactly the residual conditional-information term $I(R; X)$ of (4); and the exchangeability the basic guarantee needs is typically unavailable in deployed time-series settings. Within the class of objects it produces it is optimal, and where a guarantee is the product it is the right tool. But coverage answers a question, *is the truth in this region?*, that, in forecasting, is often not the one you were asking. Conformal prediction is not weak uncertainty quantification; it is exact uncertainty quantification for a set-valued coverage objective, and mistaking a set-coverage certificate for evidence of distributional quality is the category error.

Reproducibility. Every figure is generated by the accompanying script `figures.py` (Figures 1–7), and the time-series experiment by the benchmark scripts, all from the synthetic generators described in the text.

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